

Topic B4

Biological molecules

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B4.1	Carbohydrates, fats and proteins
B4.2	Food tests
B4.3	DNA

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Carbohydrates

Glucose is a simple reducing sugar used in respiration. Starch and glycogen are large insoluble storage carbohydrates made from glucose; cellulose forms plant cell walls. Insolubility prevents storage molecules affecting water potential and allows compact storage.

Examiner insight

Link each carbohydrate to its biological role rather than listing names only.

2. Proteins and lipids

Proteins are chains of amino acids folded into specific shapes and function as enzymes, antibodies and structural materials. Lipids contain fatty acids and glycerol, store concentrated energy, insulate and form cell membranes.

Examiner insight

A protein's function depends on its three-dimensional shape.

3. Food tests

Benedict's reagent with heat tests reducing sugar: blue to green, yellow, orange or brick-red. Iodine tests starch: orange-brown to blue-black. Biuret tests protein: blue to lilac-purple. Ethanol emulsion tests lipid: clear to milky white.

Examiner insight

Include reagent, procedure and positive colour change.

4. DNA

DNA is made of two strands coiled as a double helix. Each strand contains nucleotides with bases; base sequence stores information for amino-acid sequence and therefore protein structure.

Examiner insight

Genes are lengths of DNA, not separate materials floating in the nucleus.

Worked application

Situation: An unknown food gives lilac biuret and milky emulsion results.
Prediction: Protein and lipid are present.
Reason: Biuret identifies protein and the ethanol emulsion identifies lipid.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B4

iological molecules

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines reducing sugar? [1]
 - A a sugar that reduces Benedict's reagent when heated
 - B a polymer of amino acids folded into a specific shape
 - C a molecule formed from fatty acids and glycerol
 - D a length of DNA coding for a protein
2. Which statement best defines protein? [1]
 - A a molecule formed from fatty acids and glycerol
 - B a polymer of amino acids folded into a specific shape
 - C a length of DNA coding for a protein
 - D a repeating unit of DNA containing a base, sugar and phosphate
3. Which statement best defines lipid? [1]
 - A a length of DNA coding for a protein
 - B a repeating unit of DNA containing a base, sugar and phosphate
 - C a molecule formed from fatty acids and glycerol
 - D a sugar that reduces Benedict's reagent when heated
4. Which statement best defines gene? [1]
 - A a repeating unit of DNA containing a base, sugar and phosphate
 - B a sugar that reduces Benedict's reagent when heated
 - C a polymer of amino acids folded into a specific shape
 - D a length of DNA coding for a protein
5. Which statement best defines nucleotide? [1]
 - A a repeating unit of DNA containing a base, sugar and phosphate
 - B a sugar that reduces Benedict's reagent when heated
 - C a polymer of amino acids folded into a specific shape
 - D a molecule formed from fatty acids and glycerol
6. Which statement about carbohydrates is correct? [1]
 - A Any carbohydrate that turns iodine blue-black.
 - B Glucose is a simple reducing sugar used in respiration.
 - C A molecule made only from glycerol.
 - D A chain of glucose used only in cell walls.
7. Which statement about proteins and lipids is correct? [1]
 - A A molecule made only from glycerol.
 - B A chain of glucose used only in cell walls.
 - C Proteins are chains of amino acids folded into specific shapes and function as enzymes, antibodies and structural materials.
 - D An entire chromosome pair.

8. Which statement about food tests is correct? [1]
- A A chain of glucose used only in cell walls.
 - B An entire chromosome pair.
 - C One complete protein.
 - D Benedict's reagent with heat tests reducing sugar: blue to green, yellow, orange or brick-red.
9. Which statement about dna is correct? [1]
- A DNA is made of two strands coiled as a double helix.
 - B An entire chromosome pair.
 - C One complete protein.
 - D Any carbohydrate that turns iodine blue-black.
10. Which prediction is most appropriate when an unknown food gives lilac Biuret and milky emulsion results? [1]
- A Any carbohydrate that turns iodine blue-black.
 - B A chain of glucose used only in cell walls.
 - C Protein and lipid are present.
 - D One complete protein.

Section B · Structured questions

11 Biologists use precise vocabulary when studying biological molecules.

(a) Define reducing sugar.

[2]

(b) Define protein.

[2]

(c) Distinguish lipid from the incorrect idea: “a chain of glucose used only in cell walls”.

[2]

(d) State two important features of carbohydrates.

[2]

[Total: 8]

12 An unknown food gives lilac biuret and milky emulsion results.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records relative reducing-sugar result under three conditions.

condition	relative reducing-sugar result
sample A	0
sample B	2
sample C	4

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to compare reducing-sugar concentration in fruit juices using Benedict's reagent.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of biological molecules to an unfamiliar context.

(a) Define gene. [2]

(b) Explain the biological importance of food tests. [3]

(c) Correct this statement: “one complete protein”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. A sugar that reduces benedict's reagent when heated.
- 2 B. A polymer of amino acids folded into a specific shape.
- 3 C. A molecule formed from fatty acids and glycerol.
- 4 D. A length of dna coding for a protein.
- 5 A. A repeating unit of dna containing a base, sugar and phosphate.
- 6 B. Link each carbohydrate to its biological role rather than listing names only.
- 7 C. A protein's function depends on its three-dimensional shape.
- 8 D. Include reagent, procedure and positive colour change.
- 9 A. Genes are lengths of DNA, not separate materials floating in the nucleus.
- 10 C. Biuret identifies protein and the ethanol emulsion identifies lipid.

Solutions · Structured questions

11. (a) a sugar that reduces Benedict's reagent when heated [2]. (b) a polymer of amino acids folded into a specific shape [2]. (c) a molecule formed from fatty acids and glycerol; not a chain of glucose used only in cell walls [2]. (d) Credit any two accurate features from the guide: Glucose is a simple reducing sugar used in respiration. Starch and glycogen are large insoluble storage carbohydrates made from glucose; cellulose forms plant cell walls. Insolubility prevents storage molecules affecting water potential and allows compact storage. [2].

12. (a) Protein and lipid are present [2]. (b) Biuret identifies protein and the ethanol emulsion identifies lipid [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $4 - 0 = 4$ [2]. (b) The values change from 0 to 4, with the highest value 4. [2]. (c) A higher colour score indicates more reducing sugar when volumes and heating are controlled [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Juice sample [1]. (b) Time to first colour change or calibrated colour score [1]. (c) Same volumes, benedict's volume, temperature and heating time [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Use a colorimeter and a glucose calibration curve [1].

15. (a) a length of DNA coding for a protein [2]. (b) Award linked points from: Benedict's reagent with heat tests reducing sugar: blue to green, yellow, orange or brick-red. Iodine tests starch: orange-brown to blue-black. Biuret tests protein: blue to lilac-purple. Ethanol emulsion tests lipid: clear to milky white. [3]. (c) Correct statement: nucleotide means a repeating unit of DNA containing a base, sugar and phosphate [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.